

Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

- Terminal showing your current branch

```
PS C:\Users\dominic\source\repos\IT\IT314AB_Ocarol\my_first_flutter_app> git checkout -b feature/student-list
Switched to a new branch 'feature/student-list'
PS C:\Users\dominic\source\repos\IT\IT314AB_Ocarol\my_first_flutter_app> git branch
  feature/data-ui
* feature/student-list
  feature/widgets
  main
```

🚩 Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart `List`.

Research what a Dart `List` is and why it is useful when handling multiple pieces of similar data.

Instead of creating one student at a time with separate variables, create a collection that stores at least five students.

Each student should contain information such as:

- Image
- Name
- Course
- Year Level
- Age
- Hobby

Your data should exist separately from your UI widgets.

Screenshot:

Code:

- Your student list
- At least five student entries

```
54 List<Profile> profiles = [
55
56   Profile(
57     name: 'Dominic Ocarol',
58     course: 'BSIT 3 - 1',
59     age: 23,
60     height: 5.11,
61     image: 'assets/images/batman.jpg',
62     hobby: 'Playing Games and Watching Anime',
63     studentstatus: true,
64   ),
65
66   Profile(
67     name: 'John Doe',
68     course: 'BSIT 4 - 2',
69     age: 26,
70     height: 4.11,
71     hobby: 'Watching Anime and Playing Games',
72     image: 'assets/images/meepo.jpg',
73     studentstatus: true,
74   ),
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76   Profile(
77     name: 'Jane Smith',
78     course: 'BSIT 4 - 1',
79     age: 24,
80     height: 6.11,
81     hobby: 'Coffe Lover',
82     image: 'assets/images/meepo.jpg',
83     studentstatus: true,
84   ),
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86   Profile(
87     name: 'Michael Lee',
88     course: 'BSIT 4 - 3',
89     age: 26,
90     height: 5.09,
91     hobby: 'Coding and cooking',
92     image: 'assets/images/meepo.jpg',
93     studentstatus: true,
94   ),
95
96   Profile(
97     name: 'Nicol Cruz',
98     course: 'BSIT 1 - 1',
99     age: 18,
100    height: 4.11,
101    hobby: 'Music and Gaming',
102    image: 'assets/images/meepo.jpg',
103    studentstatus: true,
104  ),
105 ];
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```

🚩 Flag 2 — The Copy-Paste Trap

You already know how to create a Student Card.

You created five student cards manually.

You duplicated the same card design until all five students appear.

Stop.

Imagine having:

- 50 students.
- 500 students.
- 5,000 students.

Would copying the same widget over and over still be practical?

This flag is intentionally making you feel the problem before solving it.

Screenshot:

- Your five manually created student cards (do not change your past code yet)

```
134 //Profile Card 1
135 Card(
136   color: const Color.fromARGB(255, 255, 255, 255),
137   child: ListFile(
138     leading: CircleAvatar(
139       radius: 30,
140       backgroundColor: const Color.fromARGB(255, 0, 0, 0),
141       backgroundImage: const AssetImage('assets/images/batman.jpg'),
142     ), // CircleAvatar
143     title: Text(
144       profile.name,
145       style: const TextStyle(color: Colors.black),
146     ), // Text
147     subtitle: Text(
148       'Course: ${profile.course}\nAge: ${profile.age}\nHeight: ${profile.height}\nHobby: ${profile.hobby}\nStudent Status: ${profile.studentstatus ? 'Active' : 'Inactive'}',
149       style: const TextStyle(color: Colors.black),
150     ), // Text
151   ), // ListFile
152 ), // Card
153 const SizedBox(height: 20),
154
155 //Profile Card 2
156 Card(
157   color: const Color.fromARGB(255, 255, 255, 255),
158   child: ListFile(
159     leading: CircleAvatar(
160       radius: 30,
161       backgroundColor: Colors.white,
162       backgroundImage: const AssetImage('assets/images/tuglife.jpg'),
163     ), // CircleAvatar
164     title: Text(
165       profile1.name,
166       style: const TextStyle(color: Colors.black),
167     ), // Text
168     subtitle: Text(
169       'Course: ${profile1.course}\nAge: ${profile1.age}\nHeight: ${profile1.height}\nHobby: ${profile1.hobby}\nStudent Status: ${profile1.studentstatus ? 'Active' : 'Inactive'}',
170       style: const TextStyle(color: Colors.black),
171     ), // Text
172   ), // ListFile
173 ), // Card
174 const SizedBox(height: 20),
175
176 //Profile Card 3
177 Card(
178   color: const Color.fromARGB(255, 255, 255, 255),
179   child: ListFile(
180     leading: CircleAvatar(
181       radius: 30,
182       backgroundColor: const Color.fromARGB(255, 0, 0, 0),
183       backgroundImage: const AssetImage('assets/images/bard.jpg'),
184     ), // CircleAvatar
185     title: Text(
```

```

7 //Profile Card 4
8 Card(
9   color: const Color.fromARGB(255, 255, 255, 255),
10  child: ListTile(
11    leading: CircleAvatar(
12      radius: 30,
13      backgroundColor: Colors.white,
14      backgroundImage: const AssetImage('assets/images/fat.jpg'),
15    ), // CircleAvatar
16    title: Text(
17      profile3.name,
18      style: const TextStyle(color: Colors.black),
19    ), // Text
20    subtitle: Text(
21      'Course: ${profile3.course}\nAge: ${profile3.age}\nHeight: ${profile3.height}\nHobby: ${profile3.hobby}\nStudent Status: ${profile3.studentstatus ? 'Active' : 'Inactive'}',
22      style: const TextStyle(color: Colors.black),
23    ), // Text
24  ), // ListTile
25 ), // Card
26 const SizedBox(height: 20),
27
28 //Profile Card 5
29 Card(
30   color: const Color.fromARGB(255, 255, 255, 255),
31   child: ListTile(
32     leading: CircleAvatar(
33       radius: 30,
34       backgroundColor: Colors.white,
35       backgroundImage: const AssetImage('assets/images/meepo.jpg'),
36     ), // CircleAvatar
37     title: Text(
38       profile4.name,
39       style: const TextStyle(color: Colors.black),
40     ), // Text
41     subtitle: Text(
42       'Course: ${profile4.course}\nAge: ${profile4.age}\nHeight: ${profile4.height}\nHobby: ${profile4.hobby}\nStudent Status: ${profile4.studentstatus ? 'Active' : 'Inactive'}',
43       style: const TextStyle(color: Colors.black),
44     ), // Text
45   ), // ListTile
46 ), // Card
47 ], //cards
48 ), // Column
49 ), // SingleChildScrollView
50 ); // Scaffold

```

Flag 3 — Discover **ListView**

Objective: Make your student directory scrollable using **ListView**.

Research Flutter's **ListView**.

Replace your previous layout with a scrolling layout so users can browse through all of your student cards naturally.

Your application should now allow continuous vertical scrolling.

Test

The user should be able to:

- Scroll from the first student
- Continue scrolling
- Reach the last student without resizing the screen

Screenshot:

- `ListView` in your code
- Top of the application
- Bottom portion showing later students

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...profiles.map((profile) {
  return Column(
    children: [
      Card(
        color: Colors.white,
        child: ListTile(
          leading: CircleAvatar(
            radius: 30,
            backgroundColor: Colors.black,
            backgroundImage: AssetImage(
              profile.image ?? 'assets/images/meepo.jpg',
            ),
          ),
          title: Text(
            profile.name ?? 'Unknown',
            style: const TextStyle(
              color: Colors.black,
              fontWeight: FontWeight.bold,
            ),
          ),
          subtitle: Text(
            'Course: ${profile.course ?? 'Unknown'}\n'
            'Age: ${profile.age ?? 'Unknown'}\n'
            'Height: ${profile.height ?? 'Unknown'}\n'
            'Hobby: ${profile.hobby ?? 'Unknown'}\n'
            'Student Status: ${profile.studentstatus ? 'Active' : 'Inactive'}',
            style: const TextStyle(
              color: Colors.black,
            ),
          ),
        ),
      ),
    ],
  ),
  const SizedBox(height: 20),
],
);
}),
),
```

Student List

Student List

Dominic Ocarol



Course: BSIT 3 - 1

Age: 23

Height: 5.11

Hobby: Playing Games and Watching Anime

Student Status: Active

John Doe



Course: BSIT 4 - 2

Age: 26

Height: 4.11

Hobby: Watching Anime and Playing Games

Student Status: Active

Jane Smith



Course: BSIT 4 - 1

Age: 24

Height: 6.11

Hobby: Coffe Lover

Student Status: Active

Michael Lee



Course: BSIT 4 - 3

Age: 26

Height: 5.09

Hobby: Coding and cooking

Student Status: Active

Nicol Cruz



Course: BSIT 1 - 1

Age: 18

Height: 4.11

Hobby: Music and Gaming

Student Status: Active

🚩 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

Your UI should now create one card for each student in your list.

Test

Add one new student to your list.

Without creating another card manually, your application should display the new student automatically.

Screenshot:

- `ListView.builder` in your code
- Student list
- Newly generated student appearing in the app

```
Profile(  
  name: 'Ikon Cruz',  
  course: 'BSIT 2 - 2',  
  age: 18,  
  height: 4.11,  
  hobby: 'Music and Gaming',  
  image: 'assets/images/woody.jpg',  
  studentstatus: true,  
),
```



Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

Add at least three new fields to every student.

Examples:

- Student ID
- Email
- Favorite Subject

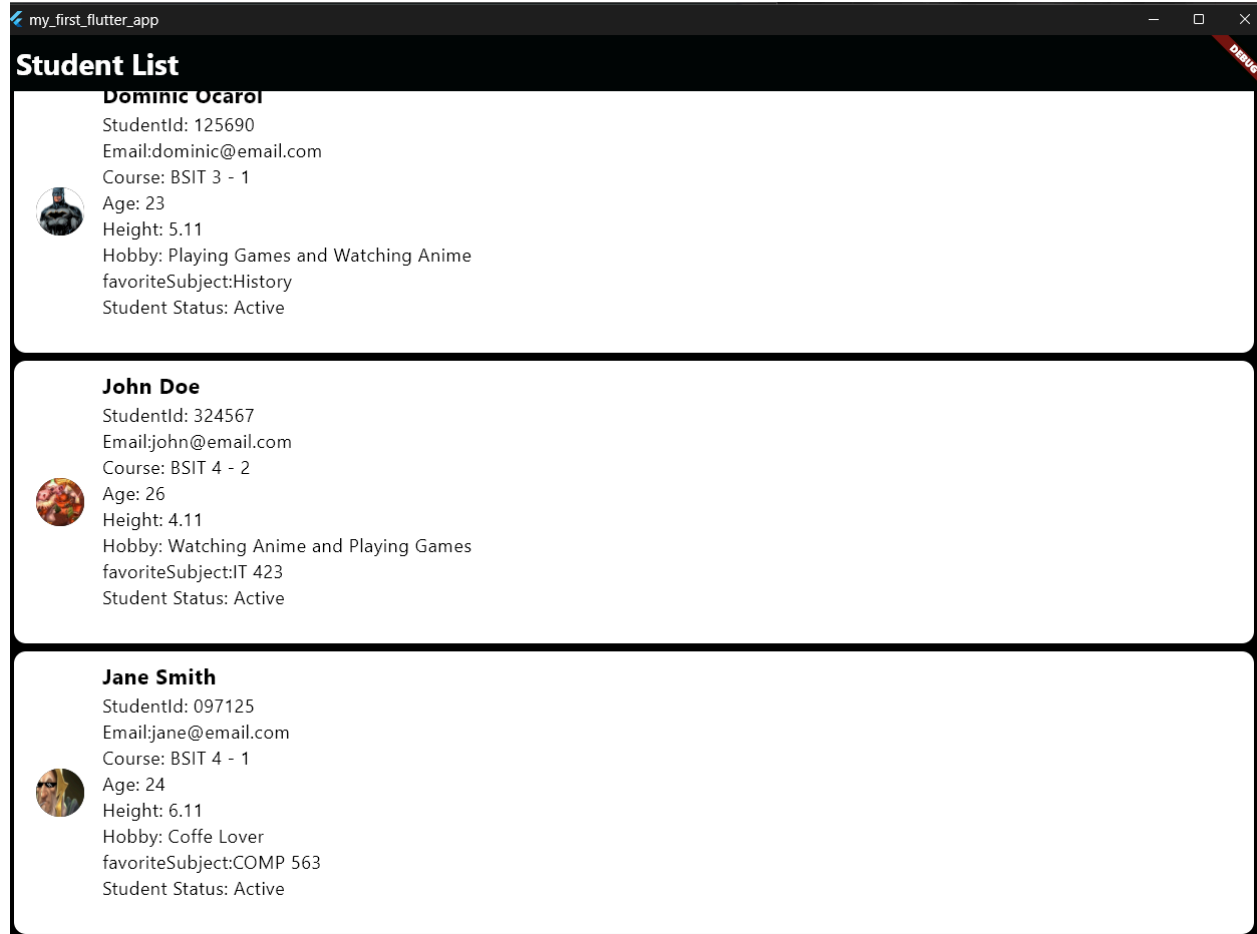
Update your Student Card so the new information appears consistently across every generated card.

Screenshot:

- Updated student data
- Student Card displaying the additional fields

```
5 class _MyHomePageState extends State<MyHomePage> {  
7   List<Profile> profiles = []  
8  
9   Profile(  
10     studentId: '125690',  
11     name: 'Dominic Ocarol',  
12     email: 'dominic@email.com',  
13     course: 'BSIT 3 - 1',  
14     age: 23,  
15     height: 5.11,  
16     image: 'assets/images/batman.jpg',  
17     hobby: 'Playing Games and Watching Anime',  
18     studentstatus: true,  
19     favoriteSubject: 'History'  
20   ),  
21  
22   Profile(  
23     studentId: '324567',  
24     name: 'John Doe',  
25     email: 'john@email.com',  
26     course: 'BSIT 4 - 2',  
27     age: 26,  
28     height: 4.11,  
29     hobby: 'Watching Anime and Playing Games',  
30     image: 'assets/images/meepo.jpg',  
31     studentstatus: true,  
32     favoriteSubject: 'IT 423'  
33   ),  
34  
35   Profile(  
36     studentId: '097125',  
37     name: 'Jane Smith',  
38     email: 'jane@email.com',  
39     course: 'BSIT 4 - 1',  
40     age: 24,  
41     height: 6.11,  
42     hobby: 'Coffe Lover',  
43     image: 'assets/images/tuglife.jpg',  
44     studentstatus: true,  
45     favoriteSubject: 'COMP 563'  
46   ),  
47  
48 };
```

```
Profile(  
  studentId: '452891',  
  name: 'Michael Lee',  
  email: 'micheal@yahoo.com',  
  course: 'BSIT 4 - 3',  
  age: 26,  
  height: 5.09,  
  hobby: 'Coding and cooking',  
  image: 'assets/images/bard.jpg',  
  studentstatus: true,  
  favoriteSubject: 'IT 203'  
)  
,  
  
Profile(  
  studentId: '023423',  
  name: 'Nicol Cruz',  
  email: 'nicol@email.com',  
  course: 'BSIT 1 - 1',  
  age: 18,  
  height: 4.11,  
  hobby: 'Music and Gaming',  
  image: 'assets/images/fat.jpg',  
  studentstatus: true,  
  favoriteSubject: 'IT 101'  
)  
,  
  
Profile(  
  studentId: '452215',  
  name: 'Ikon Cruz',  
  email: 'ikon@yahoo.com',  
  course: 'BSIT 2 - 2',  
  age: 18,  
  height: 4.11,  
  hobby: 'Music and Gaming',  
  image: 'assets/images/woody.jpg',  
  studentstatus: true,  
  favoriteSubject: 'MATH 101'  
)  
,  
  
];
```

Student List

Dominic Ocarol

StudentId: 125690

Email: dominic@email.com

Course: BSIT 3 - 1



Age: 23

Height: 5.11

Hobby: Playing Games and Watching Anime

favoriteSubject: History

Student Status: Active

John Doe

StudentId: 324567

Email: john@email.com

Course: BSIT 4 - 2



Age: 26

Height: 4.11

Hobby: Watching Anime and Playing Games

favoriteSubject: IT 423

Student Status: Active

Jane Smith

StudentId: 097125

Email: jane@email.com

Course: BSIT 4 - 1



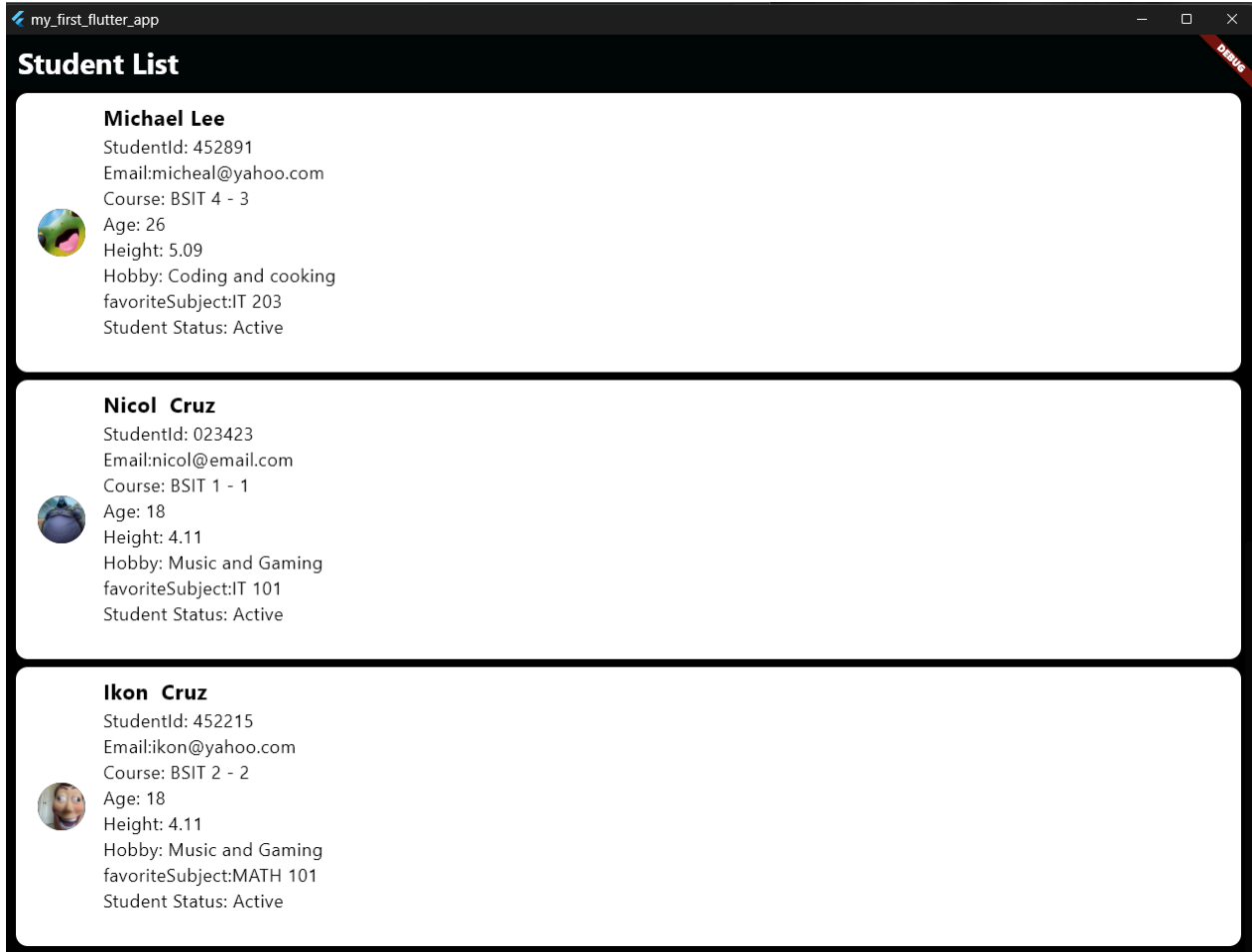
Age: 24

Height: 6.11

Hobby: Coffee Lover

favoriteSubject: COMP 563

Student Status: Active



🚩 Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty `List` gracefully.

Imagine your app just connected to a database.

The database returns:

```
[]
```

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

Examples:

- "No students found."
- "Student list is empty."

Test

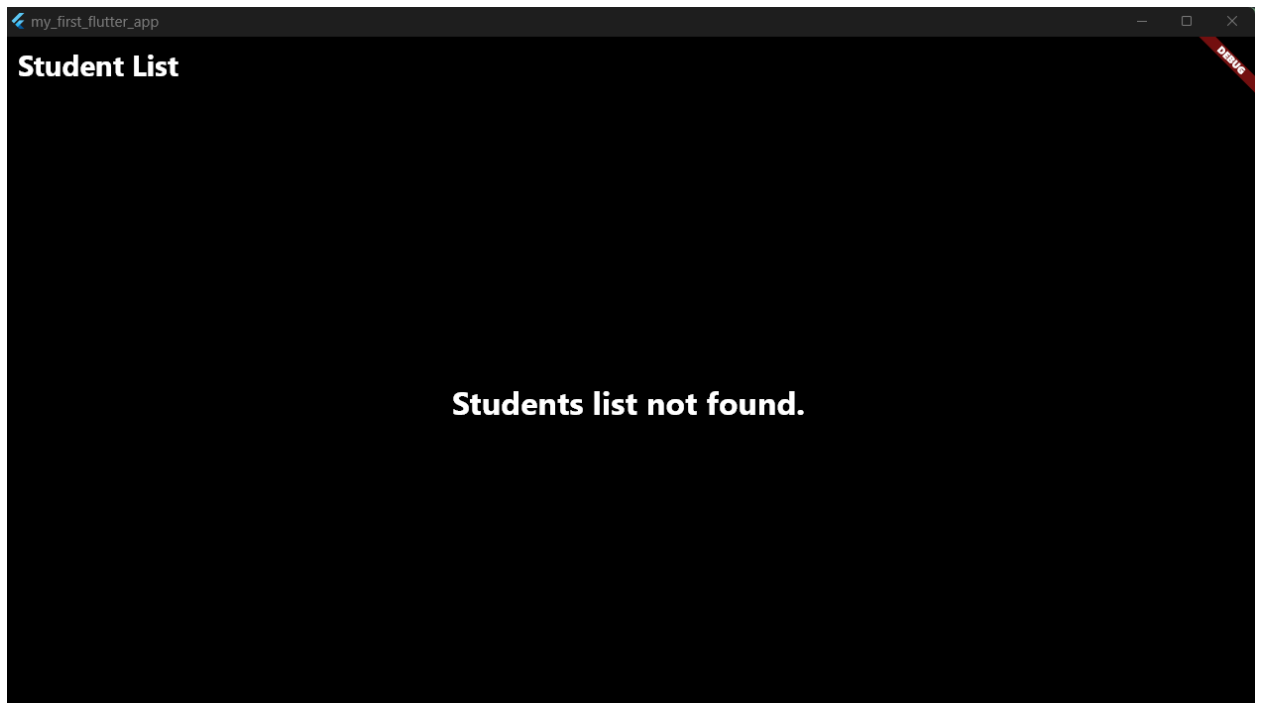
- Display your student list normally.
- Then temporarily make your list empty.
- Your application should display the empty state instead of student cards.

Screenshot:

- Empty student list

```
class _MyHomePageState extends State<MyHomePage> {  
  List<Profile> profiles = [  
      
  ];  
}
```

- Empty-state UI



- Application still functioning correctly

Flag 7 — Sort the Roll Call

Objective: Change the order of your student list.

Class lists are rarely displayed randomly.

Research how Dart can rearrange items inside a [List](#).

Sort your students alphabetically by name.

Test

Before sorting:

- Maria
- John
- Aaron
- Sophia

After sorting:

- Aaron
- John
- Maria
- Sophia

Your UI should immediately reflect the new order.

Screenshot:

- Student list before or after sorting

Student List	
	Dominic Ocarol Student ID: 125690 Email: dominic@email.com Course: BSIT 3 - 1 Age: 23 Height: 5.11 Hobby: Playing Games and Watching Anime Favorite Subject: History Student Status: Active
	John Doe Student ID: 324567 Email: john@email.com Course: BSIT 4 - 2 Age: 26 Height: 4.11 Hobby: Watching Anime and Playing Games Favorite Subject: IT 423 Student Status: Active
	Jane Smith Student ID: 097125 Email: jane@email.com Course: BSIT 4 - 1 Age: 24 Height: 6.11 Hobby: Coffee Lover Favorite Subject: COMP 563 Student Status: Active
	Michael Lee Student ID: 452891 Email: micheel@yahoo.com Course: BSIT 4 - 3 Age: 26

- Application displaying the sorted order

Student List	
	Dominic Ocarol Student ID: 125690 Email: dominic@email.com Course: BSIT 3 - 1 Age: 23 Height: 5.11 Hobby: Playing Games and Watching Anime Favorite Subject: History Student Status: Active
	Ikon Cruz Student ID: 452215 Email: ikon@yahoo.com Course: BSIT 2 - 2 Age: 18 Height: 4.11 Hobby: Music and Gaming Favorite Subject: MATH 101 Student Status: Active
	Jane Smith Student ID: 097125 Email: jane@email.com Course: BSIT 4 - 1 Age: 24 Height: 6.11 Hobby: Coffee Lover Favorite Subject: COMP 563 Student Status: Active
	John Doe Student ID: 324567 Email: john@email.com Course: BSIT 4 - 2 Age: 26

	John Doe Student ID: 324567 Email: john@email.com Course: BSIT 4 - 2 Age: 26 Height: 4.11 Hobby: Watching Anime and Playing Games Favorite Subject: IT 423 Student Status: Active
	Michael Lee Student ID: 452891 Email: micheal@yahoo.com Course: BSIT 4 - 3 Age: 26 Height: 5.09 Hobby: Coding and Cooking Favorite Subject: IT 203 Student Status: Active
	Nicol Cruz Student ID: 023423 Email: nicol@email.com Course: BSIT 1 - 1 Age: 18 Height: 4.11 Hobby: Music and Gaming Favorite Subject: IT 101 Student Status: Active

🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/student-list`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit
- ```

PS C:\Users\dominic\source\repos\IT\IT314AB_Ocarol> git commit -m "FEAT: Added Student-list"
[feature/student-list 92b59c5] FEAT: Added Student-list
 3 files changed, 172 insertions(+), 179 deletions(-)
 create mode 100644 my_first_flutter_app/assets/images/woody.jpg
PS C:\Users\dominic\source\repos\IT\IT314AB_Ocarol> git push origin feature/student-list
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 72.59 KiB | 14.52 MiB/s, done.
Total 9 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
remote:
remote: Create a pull request for 'feature/student-list' on GitHub by visiting:
remote: https://github.com/domsprog/IT314AB_Ocarol/pull/new/feature/student-list
remote:
To https://github.com/domsprog/IT314AB_Ocarol.git

```

```
PS C:\Users\dominic\source\repos\IT\IT314AB_Ocarol> git push origin feature/student-list
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 72.59 KiB | 14.52 MiB/s, done.
Total 9 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
remote:
remote: Create a pull request for 'feature/student-list' on GitHub by visiting:
remote: https://github.com/domsprog/IT314AB_Ocarol/pull/new/feature/student-list
remote:
To https://github.com/domsprog/IT314AB_Ocarol.git
 * [new branch] feature/student-list -> feature/student-list
```